Spoken Tutorial on Understanding APIs

Module	Module 3: An introduction to Building
Episode	2. Understanding APIs
Learning Objective	At the end of this tutorial learner will be able to 1. Define what an API is in the context of AI. 2. Explain the role of APIs in connecting applications to AI models. 3. Identify the purpose of API keys for authentication. 4. Describe how applications interact with APIs to send prompts and receive responses.
Approx. Duration	3-4 min
Outline	• What an API is and why it matters • Prompts as inputs sent through APIs • API keys for authentication and access control • How applications send requests and receive AI-generated responses
Meta Tags	API, AI, Python, authentication, prompts, requests, responses, models, integration, development, application, interface
Pre-requisite Tutorial	Basic understanding of Python for API interaction.

Script

Visual Cue	Narration
Title Slide	Welcome to this Spoken Tutorial on **Understanding APIs**.
Learning Objectives Slide	In this tutorial, you will learn to, • Define what an **API** is in the context of **AI**. • Explain the role of **APIs** in connecting applications to **AI** models. • Identify the purpose of **API** keys for authentication. • Describe how applications interact with **APIs** to send **prompts** and receive responses.
System Requirements Slide	Here I am using a browser on a computer or mobile.
Pre-requisite Slide	To follow this tutorial, you should have a basic understanding of Python. This includes familiarity with **API** interaction.

Illustration showing API as a messenger between two software programs, with a waiter serving food between a customer and a kitchen.	Imagine two different software programs needing to communicate. How do they exchange information and requests? This is where an **API** steps in. An **API** acts as a messenger. It allows different software programs to talk to each other. Think of a waiter in a restaurant. You give your order, a request, to the waiter. The waiter takes it to the kitchen, like an **AI** model. Then, the waiter brings back your food, the response. An **API** ensures smooth communication between applications.
Illustration showing a user typing a prompt into an application, with an arrow labeled 'API' connecting to an AI model icon.	Now, let's look at how you give instructions to an **AI** model. In **AI**, these instructions are called **prompts**. A **prompt** is a specific question or instruction you send. These **prompts** are packaged as inputs. They are then delivered to the **AI** model through an **API**. The **API** acts as the channel for your **prompts** to reach the **AI**.
Illustration showing a key icon unlocking a digital lock, symbolizing an API key providing access and authentication to AI services.	But how does an **API** know who is making a request? This brings us to **API** keys. An **API** key is a unique identifier. It authenticates your application's requests to an **API**. Consider it your digital identity. This key ensures only authorized users can access **AI** services. **API** keys control access and maintain security.
Illustration showing a complete communication cycle: an application sending a request (prompt) via an API to an AI model, and the AI model sending a response back through the API to the application.	Now, let's put it all together. How do applications fully interact with **AI** models? Applications use **APIs** to send requests. These requests contain your **prompts** for the **AI** model. The **API** delivers these **prompts** to the **AI** model. The **AI** model processes the **prompt** and generates a response. Then, the **API** relays this **AI**-generated response back. This completes the entire communication cycle. **APIs** are key for this two-way conversation with **AI** models.
Summary Slide	Let us summarize what we learned. We discussed what **APIs** are and their role as messengers. We saw how **prompts** are inputs sent to **AI** models via **APIs**. We also understood the importance of **API** keys for authentication. Finally, we covered the full request-response cycle with **AI** models.

Assignment Slide	Now as an assignment, Research three real-world applications. List how they use **APIs** to interact with **AI** models. Briefly describe the benefits of **API** integration for each. Compare the results.
EduPyramids logo	This Spoken Tutorial is brought to you by **EduPyramids Educational Services Private Limited**. **SINE**, **IIT Bombay**. Thank you for joining.